

ILLINOIS VALLEY, IL, USA

WMO: 722149

Lat: 41.352N

Lon: 89.153W

Elev: 199

StdP: 98.95

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04899

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.9	-23.5	0.5	-19.1	-21.1	0.6	-16.0	12.7	-4.5	11.8	-3.0	3.9	290	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.9	24.0	31.7	23.3	30.2	22.6	25.7	30.8	24.8	29.7	23.9	28.4	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.5	28.7	23.2	18.4	27.7	22.5	17.6	27.0	80.6	30.5	76.3	29.8	72.7	28.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.3	9.0	-23.6	35.4	5.0	2.6	-27.2	37.3	-30.2	38.8	-33.0	40.2	-36.7	42.1		
			-24.0	27.6	4.9	0.7	-27.4	28.0	-30.3	28.4	-33.0	28.8	-36.6	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-4.1	-3.0	4.2	10.9	17.2	22.2	23.7	22.5	19.4	12.4	5.1	-1.3
	DBStd	11.22	7.10	6.69	6.21	5.08	5.00	3.25	3.09	2.92	4.23	5.23	5.46	6.16
	HDD10.0	1607	439	365	199	50	4	0	0	0	0	33	165	351
	HDD18.3	3293	697	597	439	228	83	5	1	3	37	198	398	608
	CDD10.0	1916	1	2	21	76	228	367	426	388	282	107	17	1
	CDD18.3	560	0	0	2	4	48	122	169	133	69	12	0	0
	CDH23.3	5099	0	0	13	67	470	1082	1624	1102	626	114	1	0
	CDH26.7	1668	0	0	1	9	149	363	579	332	206	29	0	0
Wind	WSAvg	4.2	4.9	4.8	4.9	5.2	4.4	3.7	3.0	2.7	3.3	4.2	4.6	4.5
Precipitation	PrecAvg	869	47	40	59	77	102	97	96	98	71	74	67	42
	PrecMax	1173	96	127	122	128	193	197	228	263	148	185	145	71
	PrecMin	526	7	10	22	33	12	31	2	27	12	11	2	16
	PrecStd	146	24	28	33	32	54	54	53	58	41	46	41	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.8	25.5	27.8	32.8	34.0	36.7	33.5	33.2	30.0	22.0	15.7
		MCWB	11.7	12.2	17.9	16.8	22.2	24.1	24.9	24.5	23.3	21.7	14.7	14.3
	2%	DB	10.2	12.5	20.5	25.4	30.3	32.2	33.1	31.6	31.3	26.4	18.0	12.2
		MCWB	8.2	9.5	14.9	16.3	20.4	22.9	25.3	24.0	22.5	18.4	13.4	10.4
	5%	DB	7.4	8.4	17.2	22.8	28.0	30.6	31.4	30.0	29.0	23.7	16.0	9.1
		MCWB	5.7	6.0	12.9	14.9	18.9	22.2	24.1	23.1	21.3	17.5	12.3	7.5
	10%	DB	4.9	5.2	13.7	20.1	26.1	28.8	29.9	28.5	27.3	21.0	13.5	7.0
		MCWB	3.3	3.6	9.8	13.5	18.3	21.5	23.1	22.3	19.9	16.2	10.2	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.9	18.4	19.1	23.3	26.8	27.2	26.5	25.3	22.8	16.9	14.6
		MCDB	13.8	14.7	24.5	24.9	30.4	31.9	32.6	31.2	31.8	28.4	19.7	15.4
	2%	WB	8.6	9.6	15.7	17.4	22.0	24.5	26.0	25.3	23.7	20.6	14.5	10.4
		MCDB	9.8	13.1	19.2	22.8	27.6	29.4	31.2	30.1	28.3	24.2	16.9	11.4
	5%	WB	5.6	6.1	13.6	15.9	20.7	23.5	25.1	24.3	22.6	18.8	12.5	7.7
		MCDB	7.3	8.3	16.9	21.3	26.0	28.4	30.1	28.2	27.2	21.9	15.7	8.8
	10%	WB	3.3	3.4	10.2	14.4	19.5	22.6	24.2	23.3	21.5	16.8	10.5	5.3
		MCDB	4.9	5.0	13.4	19.5	24.3	27.4	28.8	27.1	25.8	20.3	13.5	6.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.4	9.2	11.4	11.3	10.5	10.3	10.1	11.7	10.6	8.9	7.3
		MCDBR	9.9	12.1	13.1	15.4	13.6	12.2	11.2	11.3	13.0	13.7	12.3	10.0
		MCWBR	8.2	9.0	8.5	8.6	6.4	5.7	5.4	5.4	5.7	7.2	8.3	8.6
	5% WB	MCDBR	9.8	11.3	12.0	13.6	11.7	10.1	10.2	9.6	10.7	11.0	10.8	9.5
		MCWBR	8.6	8.9	8.6	8.4	6.3	5.5	5.7	5.2	5.5	7.3	8.1	8.7
Clear-Sky Solar Irradiance	taub	0.288	0.304	0.340	0.392	0.423	0.430	0.448	0.432	0.402	0.347	0.306	0.286	
	taud	2.434	2.396	2.395	2.258	2.225	2.251	2.184	2.225	2.282	2.441	2.504	2.456	
	Ebn at Noon	870	907	909	879	855	848	827	830	830	846	845	842	
	Edh at Noon	78	94	106	131	139	136	144	134	118	90	73	70	
All-Sky Solar Radiation	RadAvg	1.72	2.64	3.65	4.68	5.41	6.13	6.10	5.39	4.55	3.02	1.97	1.45	
	RadStd	0.18	0.32	0.29	0.40	0.40	0.47	0.36	0.32	0.39	0.35	0.17	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page